The Student Research Lead Handbook

HONR 46400 · Evidence-Driven Research

Your guide to leading a class the way a researcher leads an investigation.

What a Student Research Lead is

For most Monday and Wednesday lectures this semester, one of you runs the room. That person is the **Student Research Lead**, or **SRL** for short. Your lecture slots are assigned at the start of the semester, so you will know yours early.

Being the SRL is not giving a presentation. You are not there to summarize a reading, walk through slides, or tell the class what the notebook says. You are there to run a **Socratic investigation**: a class session built out of good questions, where the room works out the ideas by thinking, committing, testing, and arguing. "Socratic" just means you lead by asking, not by telling. The best SRL sessions feel less like a lecture and more like a lab meeting where a sharp question is on the table and everyone is trying to crack it.

Here is the difference in one line. A presenter stands at the front and transfers what they know. An SRL stands at the front and makes the room think.

The philosophy: AI is your arm, not your brain

This course runs on one idea:

> **AI is your arm and your research assistant. It is not your brain.**

An arm is powerful. It reaches, fetches, lifts, and drafts. But it does not decide where to go. You do. When you lead, you model exactly that relationship for the room. You make the class think first, commit to an answer, then use AI on purpose, and then interrogate whatever it hands back. **Interrogate** here means to question a claim hard before you trust it: ask where it came from, whether it is true, and what it is quietly assuming.

Four habits sit under everything an SRL does:

1. **You make the room think.** The class does the reasoning. Your questions do the steering.
2. **You make the room commit.** Before anyone opens an AI tool, the class writes down what it believes. A **commitment** is a stated answer you put on record before you consult AI, so you can tell later whether the AI changed your mind or just agreed with you.
3. **You consult AI strategically.** You do not ask the AI to think for the room. You point it at a specific job, then bring the answer back for inspection.
4. **You interrogate what comes back.** No answer from an AI tool ends the discussion. A human decision ends the discussion.

The everyday shorthand for the whole loop is **Ask, Verify, Document**. Ask the AI for something specific. Verify what it gives you against real evidence. Document what you decided and why. You will run the room through that loop live.

The two lead formats

Each of your slots is either a Monday or a Wednesday, and each has its own shape. Both run 50 minutes. The minute frames below are fixed, and the instructor posts checkpoint signals at the section boundaries so you always know your pace.

Monday: the guided investigation

Monday is where a new idea gets built from a puzzle. The class starts from curiosity, commits to a belief, investigates with AI as the research partner, and lands on a decision it can defend.

Minutes	Section	What you do
0–9	**Your research puzzle**	Open with one concrete puzzle. Elicit prior beliefs. Get the room to commit to an answer in writing before any AI is opened.
9–31	**Guided AI investigation**	You facilitate as the class uses its AI research partner on the puzzle. You direct the prompts, run a human-versus-AI comparison, and probe what comes back. The instructor monitors for accuracy in the background.
31–43	**Human verification and formalization**	The instructor steps in to formalize the concept and lock down what is actually correct. You help connect it back to the class's committed answers.
43–50	**Decision and defense**	You close by making the room commit to a decision and defend it. The class records an AI Research Ledger line and a Claim Ticket.

Your two big blocks are **0–9** (you own it completely) and **9–31** (you run the investigation while the instructor watches accuracy). The **31–43** formalization is the instructor's block; your job there is to hand off cleanly and keep the thread visible.

Wednesday: the applied AI laboratory

Wednesday is where the idea gets used under pressure. You open with a challenge, the class works hands-on with AI, then defends the results against hard questions and carries the skill into their own projects.

Minutes	Section	What you do
0–7	**Your retrieval challenge**	Open with a challenge that forces recall and exposes a fault line. Good challenges: a common error, an ambiguous claim, a flawed AI output, a design conflict, or a result that needs interpreting.
7–30	**Applied AI laboratory**	You steer the room through hands-on work with AI. Everyone runs prompts, splits roles, and hunts for AI-failure patterns. This is the longest block, so pacing is on you.
30–38	**Peer defense**	The class defends its results while classmates ask adversarial questions. **Adversarial** here means friendly but relentless: questions designed to find the weak point, not to attack the person.
38–42	**Synthesis + accuracy lock**	You state the room's conclusion and its uncertainty in one or two lines. The instructor then locks accuracy: anything that survived challenge but is wrong gets corrected before it can enter a ledger.
42–50	**Transfer to projects**	You close by connecting the skill to each person's own research project. The class records an AI Research Ledger line and a Claim Ticket.

Your owned blocks are **0–7** and **7–30**. The **30–38** peer defense you referee; you keep the questions coming and the answers honest. At **38–42** you synthesize, and the instructor's accuracy lock has the last word before the ledger.

Your slots

Lead slots are **assigned at random at the beginning of the semester**, so you will know which lectures are yours from the first week. Nothing rotates and nothing shifts as the term goes on: your lectures are your lectures.

Which calendar meetings you drew is posted on the **course platform** in Week 1, as the slot assignment for the whole class. That posting is the single source of truth for your dates. Check it early, then look each of your dates up on the **Schedule page of the course website**, which tells you whether the meeting is a Monday or a Wednesday, which notebook it uses, and what it covers. This handbook never prints dates on purpose, so it never goes stale.

Your SRL Lead Brief

Every led lecture opens with an **SRL Lead Brief** at the very start of that lecture's notebook. It is a normal, student-visible section, so the whole class can read it. Nothing is emailed to you and nothing is held back.

Your brief names the concept in play, the compass position or design in focus, a seed puzzle you can use or adapt, and the one thing the room most needs to leave understanding.

Treat the brief as a **floor, not a ceiling**. It guarantees the session has everything it needs even on a bad week, but the good sessions are the ones where the lead adds something. Swap in an example from your own project or your own field. Restage the puzzle with different numbers, a different case, or a demonstration the room can watch. Cut a question you find flat and write a sharper one. Bring your own voice. The only fixed parts are the minute frame and the moves the rubric grades; everything else is yours to make interesting.

Your preparation timeline

Good SRL sessions are built, not improvised. Here is the rhythm.

About a week before your lead

- Read your **SRL Lead Brief** at the top of that lecture's notebook, then read the rest of the notebook and the matching chapter yourself, as a learner. You cannot lead an investigation into an idea you have not sat with.
- Pick your puzzle. Use the seed, sharpen it, or bring your own. A puzzle works when it has a real answer, more than one tempting wrong answer, and can be stated in a few sentences.
- Decide what **you** are adding that the brief does not have. One thing is enough: your own example, your own staging, your own opening question.
- Rough out your three Socratic questions and the moment you will have the room commit before touching AI.

Two days before your lead

- Submit your **preparation template** (`srl_prep_template.md`). This is your script: the puzzle, the commitment question, three Socratic questions with the answers you expect (right and wrong), the exact AI prompt or prompts the class will run, what you expect the AI to get right and wrong, your assumption-probe, your counterexample request, the decision you will make the room defend, a timing plan, and a fallback if the AI tool is down.
- The instructor reviews it and sends notes. Treat those notes seriously; they are the difference between a session that lands and one that stalls.

The day of your lead

- Arrive a few minutes early. Open Colab, confirm your AI tool responds, and paste your first prompt into a scratch cell so it is ready.
- Load your puzzle where the room can see it.
- Have your fallback ready in case the AI tool is slow or down. A dead tool should never kill your session.
- Breathe. You are not performing. You are hosting a good argument.

The ten SRL moves

These are the moves that turn a session from a talk into an investigation. You will not use all ten every time. Learn them, then reach for the ones your puzzle needs.

1. **Introduce a concrete puzzle.** Start with something specific and real, not a definition. “Here are two studies that disagree about one arrow in a diagram” beats “Today we will discuss confounding.”
2. **Ask before you define.** When a term comes up, ask the room what they think it means before you or the AI supplies the definition. Their guesses tell you where the confusion lives.
3. **Elicit prior beliefs.** Get answers on the record before any evidence arrives. “Hands up, who thinks the effect is positive?” A **prior belief** is what someone expects to be true before they look, and surfacing it makes learning visible.
4. **Require commitment before AI.** Have everyone write down their answer before the AI is opened. This is the single most important move. Without it, the AI’s answer becomes everyone’s answer and no one learns anything.
5. **Direct strategic AI use.** Point the AI at one specific job with a prompt you chose in advance. Never let the room type “explain this topic” and read the result aloud. That is outsourcing the thinking.
6. **Compare human versus AI answers.** Put the room’s committed answer next to the AI’s answer, side by side. Where they agree, ask why. Where they differ, ask who is right and how you would know.
7. **Probe assumptions.** For any claim, human or AI, ask what has to be true for it to hold. An **assumption** is a condition a claim quietly depends on. Naming it is often the whole lesson.
8. **Demand counterexamples and evidence.** Ask for a case where the claim would fail, and ask what evidence would settle it. A claim with no possible counterexample and no supporting evidence is not yet a finding.
9. **Call out unjustified AI confidence.** When the AI sounds certain, ask the room to check whether the certainty is earned. Confident tone is not evidence. This is a graded habit, not a nicety.
10. **Close by making the room defend a decision.** End on a human decision, and make someone defend it out loud against a hard question. The session is not over when the AI answers. It is over when a person commits and holds up.

When an answer is wrong or uncertain

Someone will give a wrong answer. The AI will give a wrong answer. You will not always know the right one in the moment. All three are normal, and all three are useful.

- **Treat every wrong answer as data, never as failure.** A wrong answer tells the room exactly which assumption to examine. Say “interesting, walk me through how you got there,” not “no, that’s wrong.” You are studying the reasoning, not scoring it.
- **Never humiliate.** A room that fears looking foolish stops talking, and a silent room cannot be led. Protect the person, examine the idea.
- **When you are genuinely unsure, say so and turn it into an investigation.** “I don’t know either. How could we find out?” is a strong SRL move, not a weak one. Uncertainty is the raw material of research.
- **When the AI is uncertain or hedges, notice it out loud.** Missing certainty is one of the AI-failure patterns you are hunting. Do not let a vague answer pass as a settled one.

How the instructor will step in

You are not alone at the front. The instructor is monitoring, especially during the Monday investigation block, and will step in on purpose. Expect it, and do not read it as a rescue.

- If a **conceptual error starts spreading**, the instructor flags it without taking the room from you. You keep leading; the error gets corrected.
- If the **room goes quiet**, the instructor may seed a cold-call to restart the flow. Pick the thread back up.
- If the **AI produces a failure you miss**, the instructor will often ask you to ask the room about it, rather than answering directly. Follow that thread.
- The instructor owns the **Monday 31–43 formalization block**, where the correct version of the concept gets locked in.

The full details are in the instructor intervention protocol, which you do not need to read. You just need to know that stepping in is part of the design, and it keeps your session accurate.

Logistics, in one place

- Your slots are **assigned at random at the start of the semester**, and the assignment is posted on the **course platform** in Week 1. Look your dates up on the **Schedule page** for the format and the notebook.

- Your **SRL Lead Brief** is already in the lecture notebook, at the very start. Start reading it and the notebook **about a week ahead**.
- You submit your **preparation template two days ahead**; the instructor sends notes.
- Each session ends with the class recording an **AI Research Ledger** line and a **Claim Ticket**, the two records that travel with every session in this course.
- Your leading is graded on the **SRL rubric** (`srl_rubric.md`). Read it before your first slot so you know what strong leading looks like.
- Classmates give you quick feedback after each session on the **peer feedback form** (`srl_peer_feedback_form.md`). Read it. It is the fastest way to get better before your next slot.

You are going to be nervous the first time. Every researcher is nervous the first time they run a room. The fix is not to know every answer. The fix is to ask better questions than anyone expected, and to make the room glad it had to think. Go make them think.